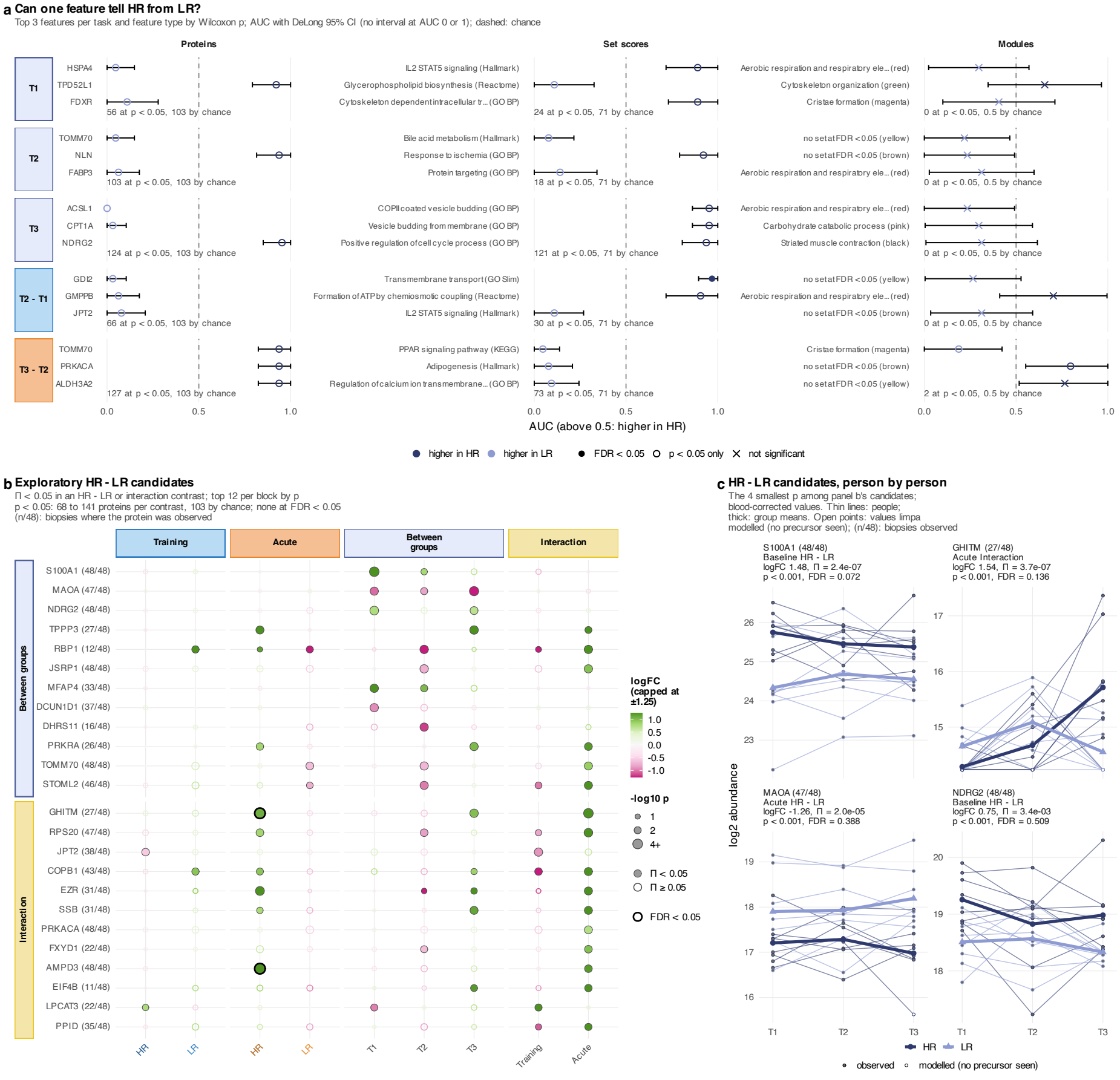

a Can one feature tell HR from LR?
Top 3 features per task and feature type by Wilcoxon p; AUC with DeLong 95% CI (no interval at AUC 0 or 1); dashed: chance
Proteins
Set scores
Modules
T1
HSPA4
TPD52L1
FDXR
56 at p < 0.05, 103 by chance
IL2 STAT5 signaling (Hallmark)
Glycerophospholipid biosynthesis (Reactome)
Cytoskeleton dependent intracellular tr... (GO BP)
24 at p < 0.05, 71 by chance
Aerobic respiration and respiratory ele... (red)
Cytoskeleton organization (green)
Cristae formation (magenta)
0 at p < 0.05, 0.5 by chance
T2
TOMM70
NLN
FABP3
103 at p < 0.05, 103 by chance
Bile acid metabolism (Hallmark)
Response to ischemia (GO BP)
Protein targeting (GO BP)
18 at p < 0.05, 71 by chance
no set at FDR < 0.05 (yellow)
no set at FDR < 0.05 (brown)
Aerobic respiration and respiratory ele... (red)
0 at p < 0.05, 0.5 by chance
T3
ACSL1
CPT1A
NDRG2
124 at p < 0.05, 103 by chance
COPII coated vesicle budding (GO BP)
Vesicle budding from membrane (GO BP)
Positive regulation of cell cycle process (GO BP)
121 at p < 0.05, 71 by chance
Aerobic respiration and respiratory ele... (red)
Carbohydrate catabolic process (pink)
Striated muscle contraction (black)
0 at p < 0.05, 0.5 by chance
T2 - T1
GDI2
GMPPB
JPT2
66 at p < 0.05, 103 by chance
Transmembrane transport (GO Slim)
Formation of ATP by chemiosmotic coupling (Reactome)
IL2 STAT5 signaling (Hallmark)
30 at p < 0.05, 71 by chance
no set at FDR < 0.05 (yellow)
Aerobic respiration and respiratory ele... (red)
no set at FDR < 0.05 (brown)
0 at p < 0.05, 0.5 by chance
T3 - T2
TOMM70
PRKACA
ALDH3A2
127 at p < 0.05, 103 by chance
PPAR signaling pathway (KEGG)
Adipogenesis (Hallmark)
Regulation of calcium ion transmembrane... (GO BP)
73 at p < 0.05, 71 by chance
Cristae formation (magenta)
no set at FDR < 0.05 (brown)
no set at FDR < 0.05 (yellow)
2 at p < 0.05, 0.5 by chance
0.0
0.5
1.0
AUC (above 0.5: higher in HR)
higher in HR
higher in LR
FDR < 0.05
p < 0.05 only
not significant
b Exploratory HR - LR candidates
Π < 0.05 in an HR - LR or interaction contrast; top 12 per block by p
p < 0.05: 68 to 141 proteins per contrast, 103 by chance; none at FDR < 0.05
(n/48): biopsies where the protein was observed
Training
Acute
Between groups
Interaction
Between groups
Interaction
S100A1 (48/48)
MAOA (47/48)
NDRG2 (48/48)
TPPP3 (27/48)
RBP1 (12/48)
JSRP1 (48/48)
MFAP4 (33/48)
DCUN1D1 (37/48)
DHRS11 (16/48)
PRKRA (26/48)
TOMM70 (48/48)
STOML2 (46/48)
GHITM (27/48)
RPS20 (47/48)
JPT2 (38/48)
COPB1 (43/48)
EZR (31/48)
SSB (31/48)
PRKACA (48/48)
FXYD1 (22/48)
AMPD3 (48/48)
EIF4B (11/48)
LPCAT3 (22/48)
PPID (35/48)
HR
LR
HR
LR
T1
T2
T3
Training
Acute
logFC (capped at ±1.25)
1.0
0.5
0.0
-0.5
-1.0
-log10 p
1
2
4+
Π < 0.05
Π ≥ 0.05
FDR < 0.05
c HR - LR candidates, person by person
The 4 smallest p among panel b's candidates;
blood-corrected values. Thin lines: people;
thick: group means. Open points: values limpa
modelled (no precursor seen); (n/48): biopsies observed
S100A1 (48/48)
Baseline HR - LR
logFC 1.48, Π = 2.4e-07
p < 0.001, FDR = 0.072
GHITM (27/48)
Acute Interaction
logFC 1.54, Π = 3.7e-07
p < 0.001, FDR = 0.136
MAOA (47/48)
Acute HR - LR
logFC -1.26, Π = 2.0e-05
p < 0.001, FDR = 0.388
NDRG2 (48/48)
Baseline HR - LR
logFC 0.75, Π = 3.4e-03
p < 0.001, FDR = 0.509
log2 abundance
T1
T2
T3
HR
LR
observed
modelled (no precursor seen)